# Akoizo Original Practice - Form 5 - Answer key

### Division B | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Which joint is usually strongest for the same glue?**

Unit: Joints, adhesives, and the bolt hook.
Worked-case data: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?
Separate investigation observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

A. A long-grain overlap with gussets

B. An end-grain butt joint

C. A joint wrapped in tape

D. A joint where glue coats the whole member

**Correct: A**

**2. (2 points) In the triangle model, which change lowers the strut compression the most?**

A. Moving the wall contact deeper (within the allowed line)

B. Making the chord thicker

C. Adding glue to the bolt hook

D. Moving the chain from 41 to 44 cm

**Correct: A**

**3. (4 points) Why does strength beyond 15 kg add nothing to the score?**

answer: The maximum Load Supported is 15,000 g, so once a structure reliably holds that, the numerator is fixed and the only way to raise the score is to reduce mass. (4 points).

**4. (2 points) When may a team adjust its structure?**

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

A. Only after the first kilogram

B. After check-in, before the six minutes

C. Before sand loading begins, within the six minutes

D. At any time

**Correct: C**

**5. (4 points) Why sort balsa by density before building?**

Unit: Wood selection, density, and buckling.
Worked-case data: A strut buckles at 120 N with an unbraced length of 30 cm. A brace is added at its midpoint. Estimate the new buckling load.
Separate investigation observations:
Stick A: 0.11 g/cm^3 - failed as a strut at 6 kg.
Stick B: 0.18 g/cm^3, same size - held 11 kg as a strut.
Stick C: 0.11 g/cm^3 with a midpoint brace - held 15 kg.

answer: Stiffness and strength rise with density and vary widely between sticks, so sorting lets you put the stiffest wood in the strut and the lightest in lightly loaded bracing. (4 points).

**6. (4 points) What two horizontal forces form the couple that balances the load moment?**

Unit: Statics of a cantilever: moments, tension, and compression.
Worked-case data: A 15 kg load hangs 42 cm from the wall. The strut meets the wall 20 cm below the bolt. Estimate the horizontal chord force.
Separate investigation observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

answer: The outward pull of the structure on the bolt (top) and the inward push of the structure against the wall at the lower contact. (4 points).

**7. (4 points) Name two items every build-log entry should record and say why.**

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

answer: Total mass and load held, because together they give the score; failure mode, because it shows what to change. (4 points).

**8. (4 points) How could you find out how much of your structure's mass is glue?**

answer: Weigh all cut pieces before assembly, then weigh the finished structure; the difference is adhesive. (4 points).

**9. (2 points) What explains the change in bracing between the designs?**

Supplied synthetic observations:
Design X: Reach 45 cm, contact 19 cm: strut needed 2 braces.
Design Y: Reach 41 cm, contact 19 cm: strut needed 1 brace.
Design Z: Reach 41 cm, contact 14 cm: strut needed 3 braces.

A. Longer reach and shallower contact both raise strut force

B. Bracing depends only on wood type

C. Shallower contact always lowers force

**Correct: A**

**10. (2 points) Define "Tier 1" in the context of Boomilever.**

answer: Placement for structures that meet every construction and competition rule. (2 points).

**11. (6 points) Solve this case and explain your reasoning: Plan the six-minute window for a team that has practiced setup.**

reasoning: 0:00-1:30 - seat the hook, check the foot clears the line, place the block on the rails.
1:30-3:00 - hang chain and bucket; confirm block above bolt plane and chain at about 41 cm.
3:00 onward - pour steadily, one partner steadies with stick tips only. (3 points).

conclusion: A rehearsed routine leaves time to pour slowly, which avoids dynamic shocks that break structures early. (3 points).

**12. (2 points) What is the best conclusion from these strut tests?**

Supplied synthetic observations:
Stick A: 0.11 g/cm^3 - failed as a strut at 6 kg.
Stick B: 0.18 g/cm^3, same size - held 11 kg as a strut.
Stick C: 0.11 g/cm^3 with a midpoint brace - held 15 kg.

A. Both density and unbraced length control buckling; bracing helped most

B. Only density matters

C. Light wood can never be a strut

**Correct: A**

**13. (2 points) Define “Tension member” in the context of Boomilever.**

answer: A member being pulled; in a Boomilever, usually the top chord. (2 points).

**14. (2 points) Define “Failure mode” in the context of Boomilever.**

answer: How a part broke: buckling, tension, joint peel, or bearing crush. (2 points).

**15. (2 points) Define “Gusset” in the context of Boomilever.**

answer: A thin plate glued across a joint to increase bonded area. (2 points).

**16. (2 points) Define “Second moment of area (I)” in the context of Boomilever.**

answer: How cross-section shape resists bending; b^4/12 for a square. (2 points).

**17. (2 points) Define “Margin” in the context of Boomilever.**

answer: Deliberate clearance from a rule limit to absorb error and deflection. (2 points).

**18. (2 points) Define “Euler load” in the context of Boomilever.**

answer: Critical buckling force, π^2EI/L^2, for a slender pinned column. (2 points).